

Recurrence Relations

Example:

the rule for finding a term is defined using preceding terms

$$a_n = a_{n-1} - a_{n-2}$$

$$a_{n-1} = a_{n-2} - a_{n-3}$$

$$a_{n-2} = a_{n-3} - a_{n-4}$$

$$a_{n-3} = a_{n-4} - a_{n-5}$$

$$\begin{array}{l}
 a_4 = a_3 - a_2 \\
 = -3 - 2 \\
 = -5 \\
 a_5 = a_4 - a_3 \\
 = -5 - (-3) \\
 = -2
 \end{array}
 \quad
 \left|
 \begin{array}{l}
 a_0 = 3 \\
 a_1 = 5
 \end{array}
 \right|
 \quad
 \begin{array}{l}
 a_2 = a_1 - a_0 \\
 = 2 \\
 a_3 = a_2 - a_1 \\
 = -3
 \end{array}$$

$$a_5 = a_4 - a_3 = (a_3 - a_2) - (a_2 - a_1)$$

$$\begin{aligned}
 &= (\cancel{a_3} - a_2) - (\cancel{a_2} - a_1) \\
 &= -a_2 + a_1 \\
 &= -5 + 3 \\
 &= -2
 \end{aligned}$$

Modelling using recurrence relations

Q. Compound interest

Initial deposit: Rs. 10,000

interest: 11% per year compounded annually

$$a_0 = 10,000$$

$$a_1 = (0.11)(a_0) + a_0 = (1.11) a_0$$

$$a_n = (1.11)^n a_0$$

$$a_0 = 10,000$$

Q. Rabbit puzzle

Initially 1 young pair of rabbits.

After 2 months, a young pair produces another pair of rabbits every subsequent month. How many pairs of rabbits are there after n months?

Soln.

$$a_0 = 1$$

$$a_1 = 1$$

$$a_2 = 2 = a_1 + a_0$$

a_i denotes no. of pairs of rabbits at the end of i months

$$a_n = a_{n-1} + a_{n-2}$$

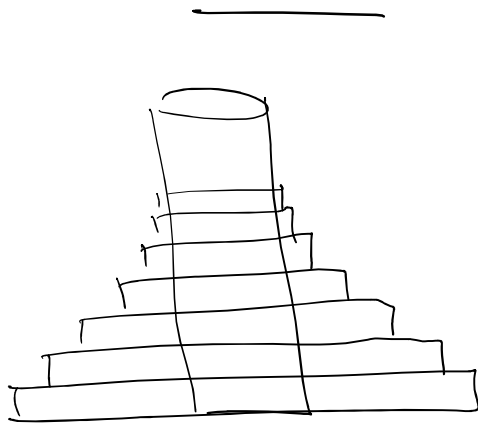
newly created pairs.

$$a_0 = 1$$

$$a_1 = 1$$

Q. Tower of Hanoi

Goal: Move everything from Peg 1 to Peg 2 from

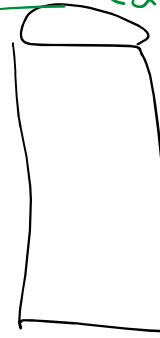


Peg 1

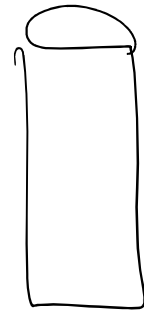
Goal: Move everything from

Peg 1 to Peg 2.

Rule: Cannot keep a larger object on top of a smaller object.



Peg 2



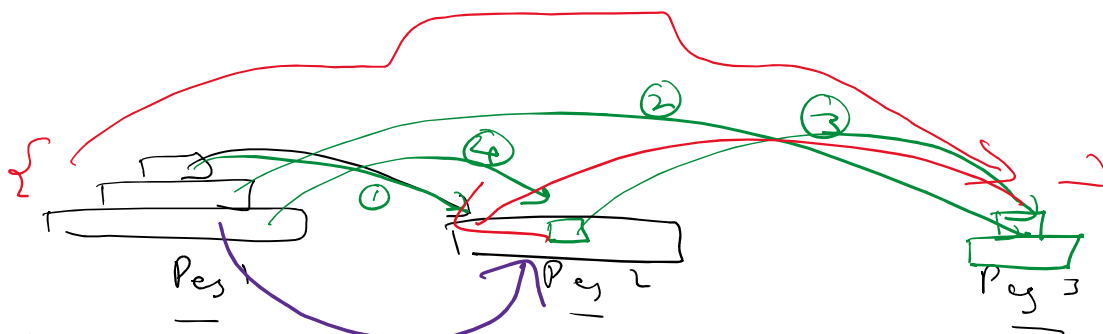
Peg 3

Let H_n denote the no. of moves needed to solve the Tower of Hanoi problem. Find a recurrence relation for H_n .

$$H_1 = 1$$

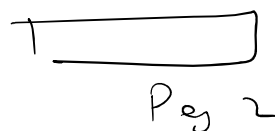
$$H_2 = 3$$

$$H_3 =$$



After 4 steps

Peg 1



Peg 2



Peg 3

$$H = H + 1 \quad | \quad H_3 = H_2 + 1 + H_2$$

$$H_3 = H_2 + 4$$

$$H_3 = H_2 + 1 + H_2$$

$$H_n = 2H_{n-1} + 1$$

$$H_1 = 1$$

$$2 \cdot 2 \cdot H_{n-2}$$

$$2^{n-1} H_1$$

Q. let $\Sigma = \{0, 1, 2, 3\}$. How many strings (with each character from Σ) of length n are there with even number of 1's?

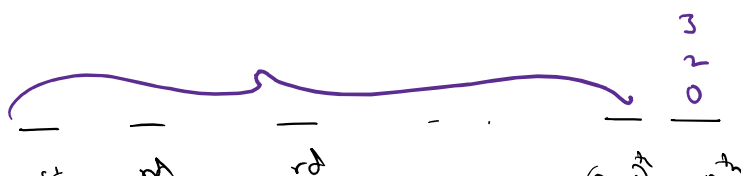
Soln. let A_n denote no. of strings of length n made of symbols from Σ that have even no. of 1's.

$$A_0 = 0$$

0, 1, 2, 3

$$A_1 = 3$$

$$A_2 = 4^2 - 6 = 10$$



$$4^2 - 6$$

$\underbrace{\quad}_{1^{st}} \quad \underbrace{\quad}_{2^{nd}} \quad \underbrace{\quad}_{3^{rd}} \quad \dots \quad \underbrace{\quad}_{(n-1)^{th}} \quad \underbrace{\quad}_{n^{th}}$

$$4^2 - 6$$

$$A_n = \underbrace{3A_{n-1}}_{\substack{n^{th} \text{ bit is} \\ \text{not 1}}} + \underbrace{4^{n-1} - A_{n-1}}_{\substack{n^{th} \text{ bit is 1.}}$$

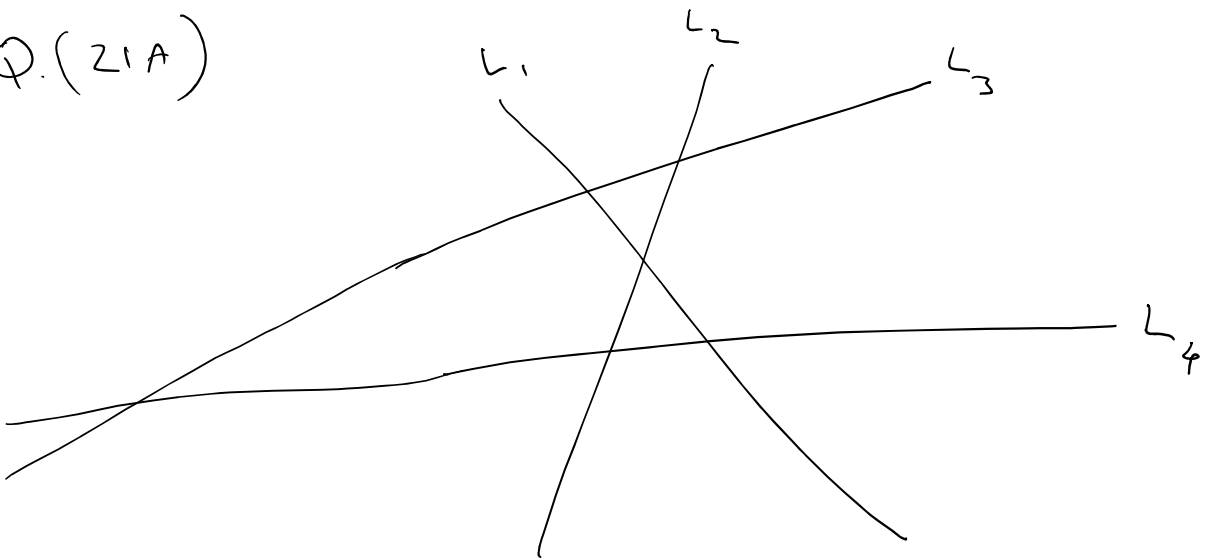
$$\boxed{A_n = 2A_{n-1} + 4^{n-1}}$$

$$A_1 = 3$$

$$A_2 = 2 \cdot 3 + 4$$

$$= 6 + 4 = 10 //$$

Q. (21A)

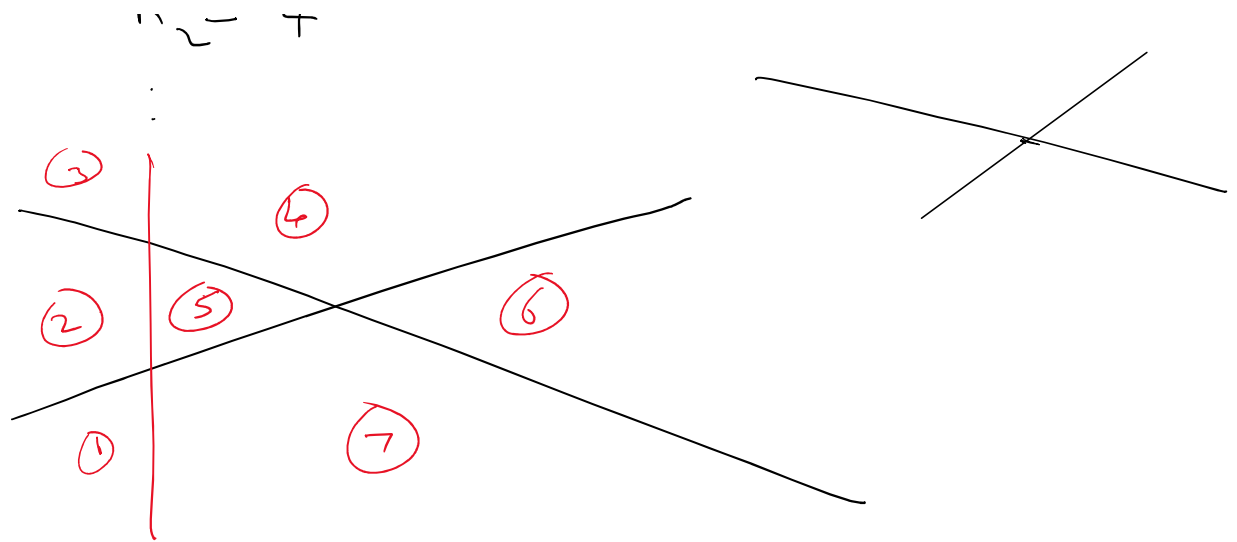


R_n : no of regions the plane is divided into by n lines.

$$R_1 = 2$$

$$R_2 = 4$$

$$R_n = R_{n-1} + n$$



Generating functions

Definition: Generating function for the sequence $a_0, a_1, a_2, \dots$ of real numbers is the infinite series

$$\begin{aligned} G(x) &= a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k + \dots \\ &= \sum_{k=0}^{\infty} a_k x^k \end{aligned}$$

Examples: Generating function of sequences $\{a_k\}$, when

(a) $a_k = 3$

$$G(x) = 3 + 3x + 3x^2 + 3x^3 + \dots$$

(b) $a_k = k+1$

$$G(x) = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$(c) \quad a_k = 2^k$$

$$\text{Then, } G(x) = 1 + 2x + 2^2 x^2 + \dots$$

Q. Generating fn for the sequence

$$1, 1, 1, 1, 1, \dots$$

$$G(x) = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$= \frac{x^5 - 1}{x - 1}$$

Q. $a_k = \binom{m}{k}$, for some +ve integer m .
What is the generating fn of the sequence

$$a_0, a_1, \dots, a_m$$

$$\begin{aligned} G(x) &= \binom{m}{0} + \binom{m}{1}x + \binom{m}{2}x^2 + \dots + \binom{m}{m}x^m \\ &= (1+x)^m \\ &= \underline{\underline{\quad}} \end{aligned}$$

Q. What sequence is the ^{simplified form} generating function

Q. What sequence is the generating function
 (a) $f(n) = \frac{1}{1-n}$ representing? simplification

Soln. $1 + n + n^2 + n^3 + \dots$

$$a_k = 1, \forall k.$$

$$(b) f(n) = \frac{1}{1-an}$$

$$1 + an + a^2 n^2 + \dots$$

$$a_k = a^k$$

Identity

$$\text{Let } f(n) = \sum_{k=0}^{\infty} a_k n^k, \quad g(n) = \sum_{k=0}^{\infty} b_k n^k$$

$$\text{Then, (i) } f(n) + g(n) = \sum_{k=0}^{\infty} (a_k + b_k) n^k \text{ and}$$

$$(ii) f(n)g(n) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) n^k$$

Q. What sequence is the generating function

Q. What sequence is the generating function $f(x) = \frac{1}{(1-x)^2}$ representing.

Soln. $f_1(x) = \frac{1}{1-x}$, $f_2(x) = \frac{1}{1-x}$

So, $f(x) = f_1(x) f_2(x)$

$$f_1(x) = f_2(x) = 1 + x + x^2 + x^3 + \dots$$

Using the above identity,

$$f(x) = f_1(x) f_2(x) = \sum_{k=0}^{\infty} \binom{k+1}{k} x^k.$$

Extended Binomial Coeff.

Let u be a real number and k be a non-negative integer. Then, the extended Binomial coefficient,

$$\binom{u}{k} = \frac{u(u-1)(u-2)\dots(u-k+1)}{k!}, \text{ if } k > 0$$

$$= 1, \text{ if } k = 0$$

Examp₁.

Example.

$$\binom{-2}{3} = \frac{(-2)(-2-1)(-2-2)}{3!}$$

$$\binom{1/2}{3} = \frac{(1/2)(1/2-1)(1/2-2)}{3!}$$

Let n be a true integer. Then,

$$\binom{-n}{r} = \frac{-n(-n-1)(-n-2) \dots (-n-r+1)}{r!}$$

$$= \frac{(-1)^r n(n+1)(n+2) \dots (n+r-1)}{r!}$$

$$= (-1)^r \binom{n+r-1}{r}$$

So $\boxed{\binom{-n}{r} = (-1)^r \binom{n+r-1}{r}, \text{ where } n \text{ is a true integer.}}$

Extended Binomial Theorem

Let n be a real number with $|x| < 1$

Let n be a real number n

and let $|x| < 1$

and let u be a real number. Then,

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$

Note: When u is a true integer, it gives the Binomial Thm.

Q. What sequence is the generating fn representing?

$$(a) (1+x)^{-n}$$

By Extended Binomial Thm,

$$\begin{aligned} (1+x)^{-n} &= \sum_{k=0}^{\infty} \binom{-n}{k} x^k \\ &= \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k \end{aligned}$$

$$\begin{aligned} (b) (1-x)^{-n} &= \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} (-x)^k \\ &= \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k \end{aligned}$$

Application of generating functions in counting problems

Q. Find the no. of solutions of

$$e_1 + e_2 + e_3 = 17$$

where we want

$$2 \leq e_1 \leq 5,$$

$$3 \leq e_2 \leq 6, \text{ and}$$

$$4 \leq e_3 \leq 7$$

$$g(x) = (x^2 + x^3 + x^4 + x^5) (x^3 + x^4 + x^5 + x^6) (x^4 + x^5 + x^6 + x^7)$$

coeff of x^{17} of $g(x)$ is
the answer.

Q. In how many different ways can 8 identical cookies be distributed among 3 children if each receives at least 2 cookies and no more than 4 cookies?

Soln.

$$g(x) = (x^2 + x^3 + x^4) (x^2 + x^3 + x^4) (x^2 + x^3 + x^4)$$

coeff. of x^8 in $g(x)$ is the answer.

Q. Use generating fn to determine the no. of ways to insert tokens worth Rs. 1, Rs. 2, Rs. 5 into a vending machine to pay for an item that costs 'r' rupees?

(a) order of tokens does not matter.

$$\underbrace{(1 + x + x^2 + x^3 + \dots)}_{\text{no. of 1 rupee coins used}} \underbrace{(1 + x^2 + x^4 + x^6 + \dots)}_{\text{no. of Rs. 2 coins used}} \underbrace{(1 + x^5 + x^{10} + x^{15} + \dots)}_{\text{\# Rs. 5}}$$

(b) if order matters

→ if you are using ^{exactly} k tokens

coeff of x^r in $(x + x^2 + x^5)^k$

we don't know k?

k = 3

same $\begin{cases} 2, 2, 5 \\ 2, 2, 5 \end{cases}$
 $\downarrow$
 2 5 2

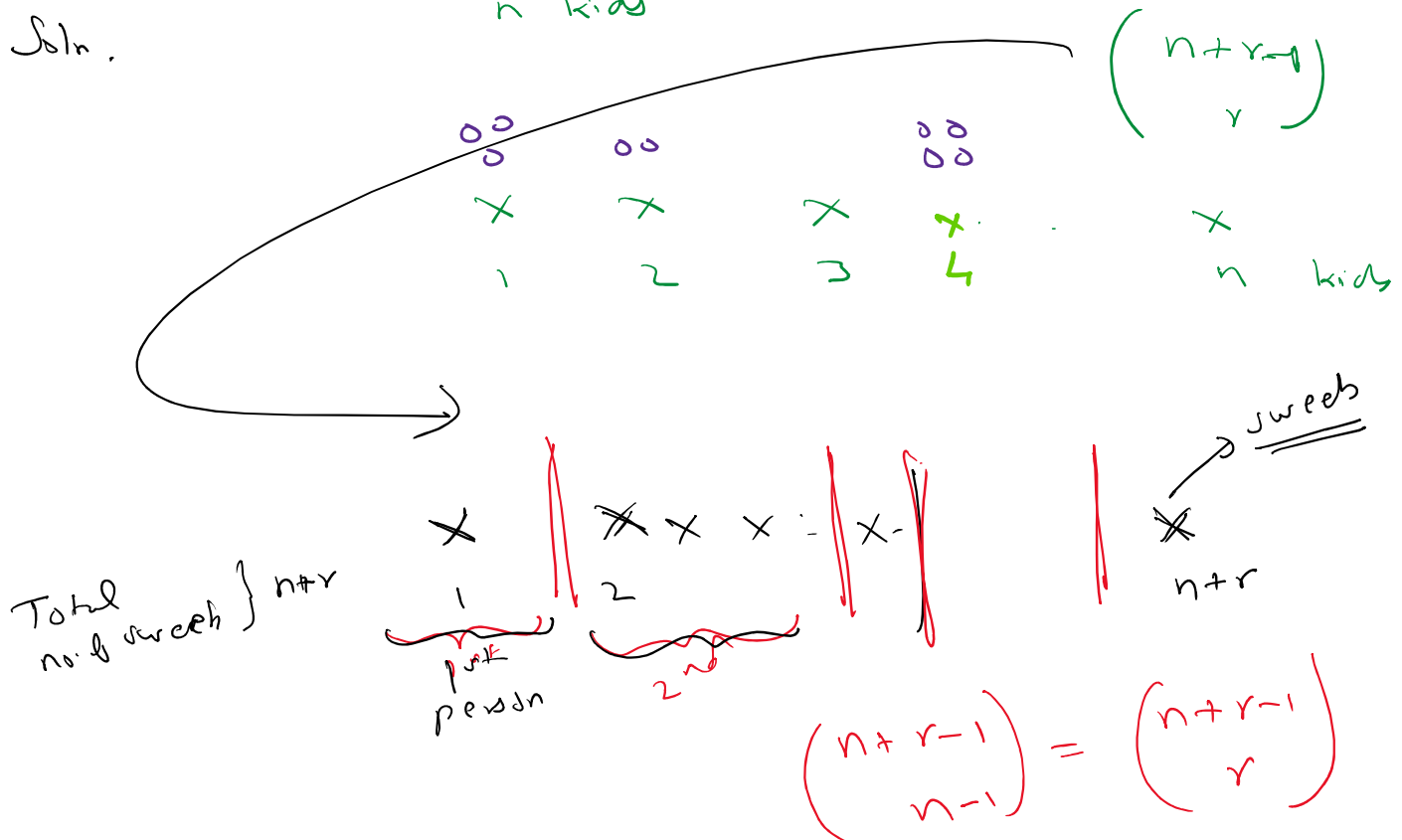
$$1 + (x + x^2 + x^5) + (x + x^2 + x^5)^2 + (x + x^2 + x^5)^3 + \dots$$

coeff. of x^r in this.

Q. How many ways to choose r sweets items from a set of n items, when repetition is allowed

a set of n items, when repetition is allowed
 or n kids

Soln.



modelling using generating fns.

$$\left(1 + x + x^2 + x^3 + \dots\right)^n = \underbrace{\left(1 + x + x^2 + \dots\right)}_{\substack{\downarrow \\ \# \text{ sweets} \\ \text{1st kid} \\ \text{gen}}} \underbrace{\left(1 + x + x^2 + \dots\right)}_{\substack{\downarrow \\ \text{2nd} \\ \text{kid}}} \dots$$

coeff of x^r in $\dots$

$$= \frac{1}{(1-x)^n}$$

coeff of x^r in $\frac{1}{(1-x)^n}$ is

$$(-1)^r \binom{-n}{r} = (-1)^r (-1)^r \binom{n+r-1}{r}$$

$$= \binom{n+r-1}{r}$$

Q. How many ways to select r objects from n different types if we must select at least one object of each type.

Soln. $\left(x + x^2 + x^3 + \dots\right)^n$
 coeff. of x^r in $\uparrow$

Balanced Parenthesis

Let A_n denote no. of ways to parenthesize

$$x_0 x_1 x_2 \dots x_n =$$

$$A_1 = 1$$

$$A_2 = 2$$

$$A_3 = 5$$

$$A_n =$$

$$x_0 x_1 \dots x_i \dots x_n$$

(2)

$$x_0, x_1, \dots, x_i, \dots, x_n, x_n$$

$$A_n = \sum_{i=0}^{n-1} A_i A_{n-i-1}$$